

Chapter 5

Myths and Realities of Lumbar Spine Stability

Stability is a popular term in discussions of the low back, but it may be widely misunderstood and inappropriately used. It has instigated passionate debate, and I have commented on some perceived biases and misunderstandings (McGill, 2011, 2013; Hodges et al., 2013b). In previous chapters we established several relevant facts. First, all sorts of tissue damage result in joint laxity, which in turn can lead to instability. For example, strained or failed ligaments cause joint laxity and unstable motion under load. End-plate fractures with loss of disc height are another example of tissue damage that results in unstable joint behavior. Clearly, joint instability is a consequence of tissue damage (this is nicely summarized by Oxland et al., 1991). A fundamental tenet is that lost mechanical integrity in any load-bearing tissue will result in stiffness losses and an increased risk of unstable behavior. We also saw in chapter 4 that during an event in which instability was observed (the buckling spine of the powerlifter), injury resulted. So, instability can both cause and be the result of injury. Finally, overlaying the tissue-based aspects of stability are the motor control aspects because coordinated contraction stiffens the joints and ultimately determines joint stability.

This chapter provides a definition of stability and an understanding of how it is increased or decreased. This is a critical foundation for those prescribing stabilization exercise or recommending strategies to prevent injury. Attempts to enhance stability and prevent instability are compromised without an understanding of the influencing factors. To quantify those factors, however, we must agree on definitions. What exactly do we mean when we use the terms spine stability, core stability, and stabilization exercise? Often, the meanings depend on the background of the person. To the biomechanist the terms pertain to a mechanical structure that can become unstable when a critical point is reached; a surgeon may view abnormal joint motion patterns as unstable but correctable through changing the anatomy; and the manual medicine practitioner may interpret patterns of muscle

coordination and posture as indicative of instability and attempt to alter one, or a few, muscle activation profiles. Several groups have made contributions to the stability issue, but only a very few have attempted to actually quantify stability. This critical issue is addressed here.

Upon completion of this chapter, you will understand stability and its importance in injury prevention and rehabilitation. Furthermore, you will understand why certain approaches are preferable for achieving sufficient stability.

Why Spine Stability Is Important for Everyone

Why do so many feel so passionately about spine stability, core stability, and stabilization exercise? Core training, abdominal training, and core stiffness and stability are all essential components for pain control, performance enhancement, and injury resilience. The explanation has four components: (1) Proximal stiffness (meaning the lumbar spine and core) enhances distal athleticism and limb speed; (2) a muscular guy wire system is essential for the flexible spine to successfully bear load; (3) muscular coactivation creates stiffness to eliminate micromovements in the joints that lead to pain and tissue degeneration; and (4) abdominal armor is necessary for people in some occupational, combative, and impact situations.

How does core stiffness enhance limb speed and strength? Consider an example with the pectoralis major (pec) muscle, which attaches to the rib cage at its proximal end, crosses the shoulder joint, and attaches to the humerus at its distal end. When muscles contract, they try to shorten. Consider the specific action here: the arm flexes around the shoulder joint moving the arm from muscle shortening. But the same shortening also bends the rib cage toward the arm at the proximal end of the muscle. Thus, using just the pec muscle would not result in a fast or forceful push or punch. If the proximal end of the pec muscle attachment were stiffened, however—that is, the rib cage couldn't move—100% of pec muscle shortening would be directed to action at its distal end, producing fast and forceful motion in the arm. In the same way a stiffened core locks down the proximal ends of the hip muscles, thus producing faster leg motion. A loss of core stiffness causes the torso to bend when sprinting, resulting in a loss of speed: some force was robbed that should have been expressed in leg velocity. The same occurs during lifting, stair climbing, carrying, and chopping firewood. Thus, a universal law of human movement is established: Proximal stiffness enhances distal mobility and athleticism. This requires core stiffness training, not movement.

The second reason core training is essential is due to the fact that the spine is a stack of vertebrae that is called upon to bear loads, yet it is flexible. An engineer cannot design a structure to be good at both. A steel beam that is straight and stood on its end is stiff and can bear loads that try to compress, shear, and twist it. The beam can bear load, but it can't move. A flexible rod

that allows movement will bend and buckle under load, but absorbs shock. Our spines do it all—they bend and allow the lungs to fill with air, and even allow us to dance. The spine is this beautiful structure that is flexible and allows flowing movement, but it requires a three-dimensional guy wire system to stiffen and stabilize it when it is required to bear loads. An analysis of the muscular system, together with its associated fascia sheets, reveals a clever guy wire system that creates balanced stiffness, eliminating the possibility of buckling and injury. The concern is that modern living does not tune and train this guy wire system. In many people it lapses into complacency. This requires core stiffness training, not movement.

The third reason core work is so important is that back injury causes tissue and joint laxity. Injury to the disc causes it to lose height, allowing aberrant micromovements. The micromovements irritate sensory nerves, which results in back pain and radiating pains. Spine stiffness from cocontracted torso muscles minimizes micromovements and controls pain. Again, this requires core stiffness training, not movement.

Fourth, some workers (e.g., fishermen, oil roughnecks, soldiers, firefighters) and combative and impact athletes require abdominal armor. Top combative athletes seek my consulting expertise for back pain. Typically, they have trained high repetition sit-ups to build armor, but eventually develop back pain that ends their careers. I change their flexion movement approach to flexion moment by having them perform exercises such as stir the pot (see chapter 11). No motion occurs in the spine. As a result, their ability to train pain-free is restored, and their careers are salvaged.

Finally there is practical, or applied, evidence from group trials regarding the value of core training. For example, military groups have made speed sit-ups mandatory for annual fitness tests. Recognizing their unacceptable low back injury statistics, some replaced sit-ups with planks and the stir the pot exercise, which are consistent with core stiffening training (not movement) (Childs et al., 2010). The groups using moment training performed better in the sit-up tests than the groups training with sit-ups, even though they did not train them. The reason is that they had healthier backs.

The big three exercises described in this book have been proven to enhance stiffness (Lee and McGill, 2015), tune the guy wire support loads, and eliminate the micromovements that cause pain (Ikeda and McGill, 2013).

Stability: A Qualitative Analogy

The following demonstration of structural stability illustrates key issues. Suppose a fishing rod is placed upright and vertical with the butt on the ground. If the rod were to have a small load placed in its tip (e.g., 1-2 lb, or 0.5-0.9 kg), it would soon bend and buckle. Now suppose that the same rod has guy wires attached at various levels along its length and that those wires are also attached to the ground in a circular pattern (see [figure 5.1](#), a and b). Each guy wire is pulled to the same tension (this is critical). Now if the tip of the rod is loaded as before, the rod can sustain the compressive forces. If you reduce the tension in just one of the wires, the rod will buckle at a reduced load; we could actually predict the node, or locus, of the buckle.

Figure 5.1 (a) The spine is analogous to a fishing rod placed upright with the butt on the ground. When compressive load is applied downward to the tip, it buckles quickly. (b) Attaching guy wires at various levels and in different directions and, most important, tensioning each guy wire to the same tension ensures stability even with massive compressive loads. Note that the guy wires need not have high tension forces, but that the tensile forces must be of roughly equal magnitude. This is the role of the musculature in ensuring sufficient spine stability.



Photos from Stuart McGill

Compressive loading similar to this has been performed on human lumbar spines. Typically, an osteoligamentous lumbar spine from a cadaver with muscles removed (and no guy wires) will buckle under approximately 90 N of compressive load (first noted by Lucas and Bresler, 1961). This is all a spine can withstand!

This analogy demonstrates the critical role of the muscles (the guy wires) to first ensure sufficient stability of the spine so that it is prepared to

withstand loading and sustain postures and movement. Also demonstrated with this example is the role of the motor control system, which ensures that the tensions in the cables are proportional so as to not create a nodal point at which buckling will occur. Revisiting the buckling injury that we observed fluoroscopically in the powerlifter ([figure 4.24](#) in chapter 4), we would hypothesize that it was caused by a motor control error in which possibly one muscle reduced its activation or, from the previous analogy, lost its stiffness. The synchrony of balanced stiffness produced by the motor control system is critical. Now we can address how stability is quantified and modulated.

Quantitative Foundation of Stability

This section quantifies the notion of stability from a spine perspective. During the 1980s, Professor Anders Bergmark of Sweden very elegantly formalized stability in a spine model with joint stiffness and 40 muscles (Bergmark, 1987). In this classic work he was able to represent mathematically the concepts of energy wells, stiffness, stability, and instability. For the most part, this seminal work went unrecognized largely because the engineers who understood the mechanics did not have the biological or clinical perspective, and clinicians were hindered in the interpretation and implications of the engineering mechanics. This section synthesizes Bergmark's pioneering effort as well as its continued evolution in the work of several others and attempts to encapsulate the critical notions without mathematical complexity. If you are mathematically inclined, see Bergmark's original work or its formalization by Cholewicki and McGill (1996).

Potential Energy as a Function of Height

The concept of stability begins with potential energy, which for our purposes here is of two basic forms. In the first form, objects have potential energy (PE) by virtue of their height above a datum.

$$\text{PE} = \text{mass} \times \text{gravity} \times \text{height}$$

Critical to measuring stability are the notions of energy wells and minimal potential energy. A ball in a bowl is considered stable because if a force (or a perturbation) were applied to it, it would rise up the side of the bowl but then come to rest again in the position of least potential energy at the bottom of the bowl (the energy well) (see [figure 5.2](#), a-d). As noted by Bergmark, “stable equilibrium prevails when the potential energy of the system is minimum.” The system is made more stable by deepening the bowl or by increasing the steepness of the sides of the bowl, or both (see [figure 5.3](#)). Thus, the notion of stability encompasses the unperturbed energy state of a system and a study of the system following perturbation. If the joules of work done by the perturbation are less than the joules of potential energy

inherent to the system, then the system will remain stable (i.e., the ball will not roll out of the bowl). The corollary is that the mechanical system will become unstable and possibly collapse if the applied load exceeds a critical value (determined by potential energy and stiffness).

Figure 5.2 The continuum of stability. (a) The deepest bowl is most stable, and (d) the hump is least stable. The ball in the bowl seeks the energy well, or position of minimal potential energy ($m \cdot g \cdot h$). Deepening the bowl or increasing the steepness of the sides increases the ability to survive perturbation. This increases stability.

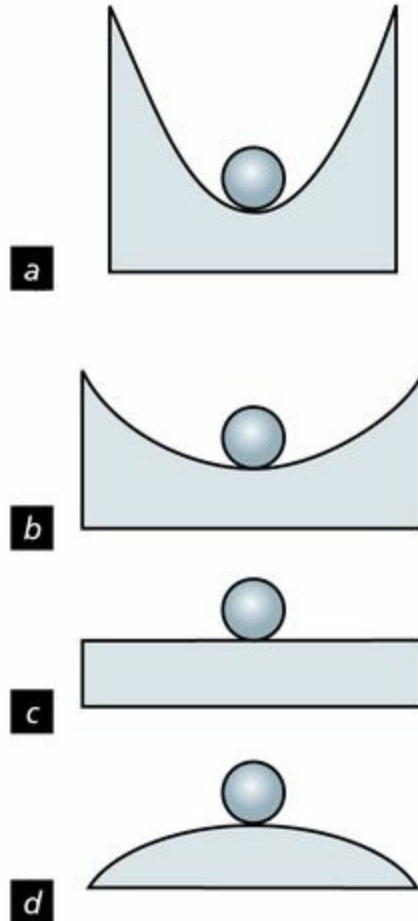
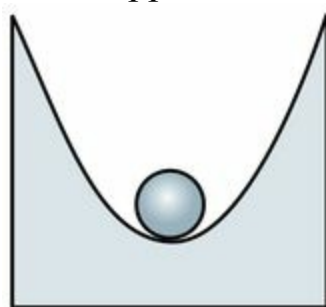


Figure 5.3 The steepness of the sides of the bowl corresponds to the stiffness of the passive tissues of the joint, which creates the mechanical stop to motion. The width of the bottom of the bowl corresponds to joint laxity. For example, a positive drawer test of the knee would be represented by a flattened bottom of the curve in which small applied forces produce large unopposed motion.



$$PE = m \cdot g \cdot h$$

Slope = joint stiffness

Width at bottom of bowl = joint laxity

The previous ball analogy is a two-dimensional example. This would be analogous to a hinged skeletal joint that has the capacity only for flexion–extension. Spinal joints can rotate in three planes and translate along three axes, requiring a six-dimensional bowl for each joint. Mathematics enables us to examine a 36-dimensional bowl (6 lumbar joints with 6° of freedom each) representing the whole lumbar spine. If the height of the bowl were decreased in any 1 of these 36 dimensions, the ball could roll out. In clinical terms, a single muscle having an inappropriate force (and thus stiffness) or a damaged passive tissue that has lost stiffness can cause instability that is both predictable and quantifiable.

Some clinicians have confused spine stability with whole-body balance and stability, which involves the center of mass and base of support in the context of falling over; this is quite different from spine stability. [Figure 5.4](#), a and b, illustrates the mechanics of whole-body balance and stability. [Figure 5.5](#), a and b, shows the clinical practice of prodding patients. This approach is misguided for enhancing spine stability.

Figure 5.4 Another type of stability, often confused with spinal column stability, involves the center of mass and base of support in the context of falling over. (a) This triangle is stable because a small perturbation to its top would not cause it to fall. The size of this stability can be quantified, in part, by the size of the angle θ . (b) As the center of the mass approaches a vertical line drawn from the base of support, a smaller perturbing force would be required for it to fall, demonstrating that it is stable but less so.

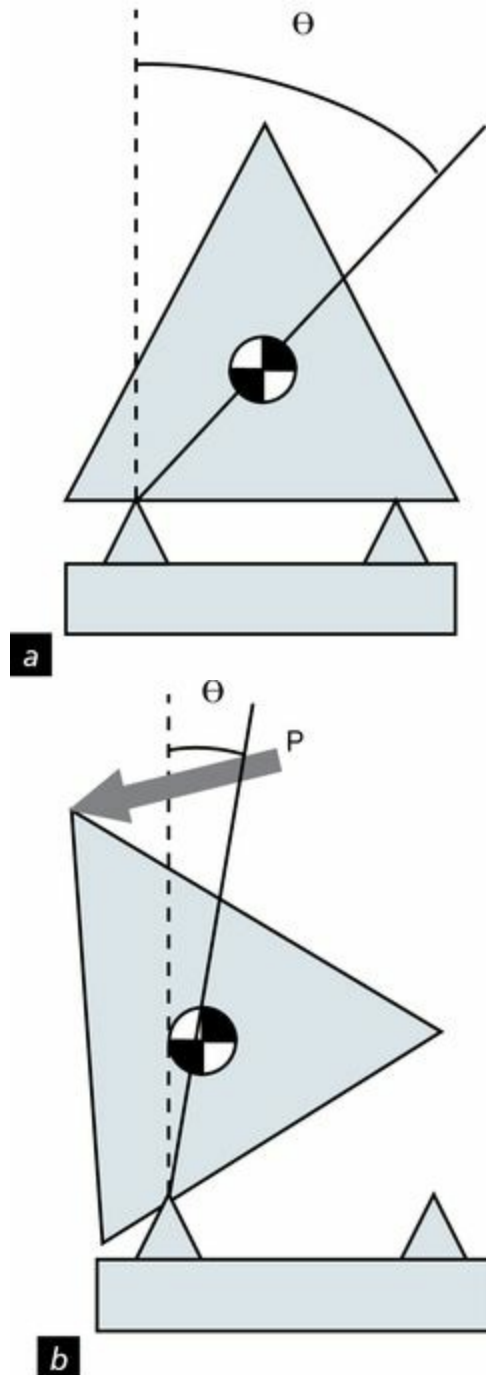


Figure 5.5 (a, b) Misunderstanding stability, illustrated in the examples given in figure 5.4, has led clinicians to try to enhance spine stability by prodding patients and attempting to knock them off balance. This is not spine stability but whole-body stability.



Photos from Stuart McGill

Potential Energy as a Function of Stiffness and Elastic Energy Storage

Although potential energy by virtue of height is useful for illustrating the concept, potential energy as a function of stiffness and storage of elastic energy is actually used for musculoskeletal application. Elastic potential energy is calculated from stiffness (k) and deformation (x) in the elastic element, as shown here:

$$PE = \frac{1}{2} \times k \times x^2$$

We will use an elastic band as an example. Stretching the band with a stiffness (k) and a distance x, it will store energy (PE). The stretched band has the potential to do work when stretched. In other words, the greater the stiffness (k), the greater the steepness of the sides of the bowl (from the previous analogy), and the more stable the structure. Thus, stiffness creates stability (see [figure 5.6](#)). More specifically, symmetrical stiffness creates even more stability. (Symmetry in stiffness is achieved by virtually all muscles of the torso; see [figure 5.7](#), a and b.) If more joules of work are performed on the spine than there are joules of potential energy due to stiffness, the spine will become unstable (see [figure 5.8](#)).

Figure 5.6 Increasing the stiffness of the cables (muscles) increases the stability (or deepens the bowl) and increases the ability to support larger applied loads (P) without failing. But most important, ensuring that the stiffness is balanced on each side enables the column to survive perturbation from either side. Increasing stiffness of just one spring will actually decrease the ability to bear compressive load (lowering PE in one direction and mode), illustrating the need for balancing stiffness for a given posture and moment demand.

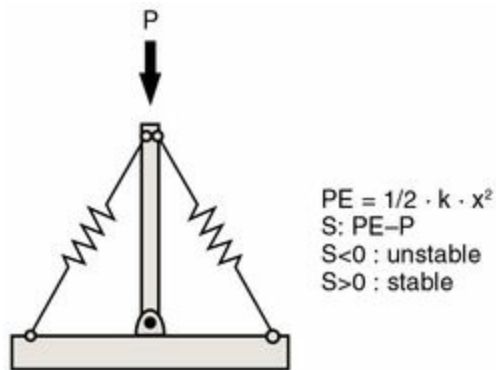


Figure 5.7 (a) Spine stiffness (and stability) is achieved by a complex interaction of stiffening structures along the spine and (b) those forming the torso wall. Balancing stiffness on all sides of the spine is more critical to ensuring stability than having high forces on a single side.

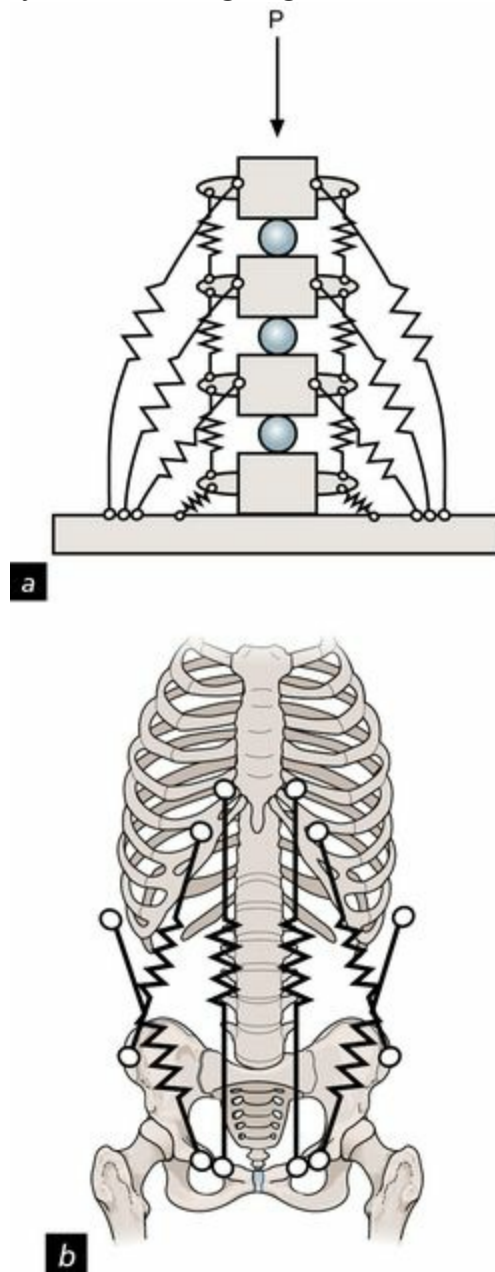
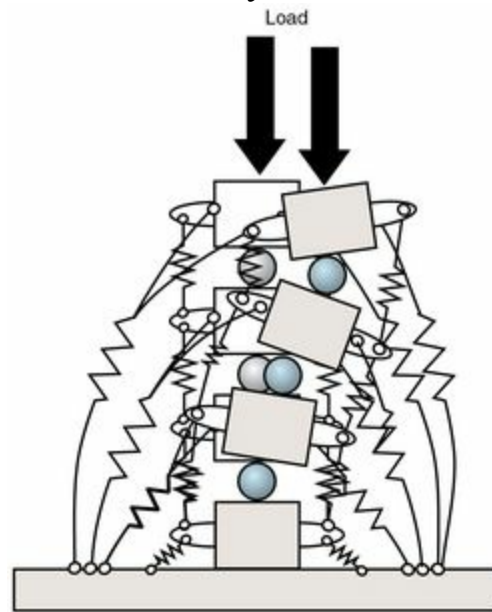


Figure 5.8 If more joules of work are performed on the spine than there are joules of potential energy due to stiffness, the spine will become unstable. In this case, the instability mode is buckling.



Active muscle acts like a stiff spring; and in fact, the greater the activation of the muscle, the greater this stiffness. Hoffer and Andreassen (1981) showed that joint stiffness increases rapidly and nonlinearly with muscle activation such that only very modest levels of muscle activity create sufficiently stiff and stable joints. Brown and McGill documented the presence of this phenomenon in the muscularly stiffened spine (Brown and McGill, 2008, 2009, 2010). Furthermore, joints possess inherent joint stiffness because the passive capsules and ligaments contribute stiffness, particularly at the end range of motion. The motor control system is able to control the stability of the joints through coordinated muscle coactivation and to a lesser degree by placing joints in positions that modulate passive stiffness contribution. However, a faulty motor control system can lead to inappropriate magnitudes of muscle force and stiffness, allowing a valley for the ball to roll out or, clinically, for a joint to buckle or undergo shear translation. However, we are limited in our ability to analyze the local stability of mechanical systems and particularly musculoskeletal linkages, because the energy wells are not infinitely deep and the many anatomical components contribute force and stiffness in synchrony to create surfaces of potential energy in which many local wells exist. Thus, we locate local

minima by examining the derivative of the energy surface. (See Bergmark, 1987; Cholewicki and McGill, 1996; and Ikeda and McGill, 2013, for mathematical detail.)

We quantify spine stability, then, by forming a matrix in which the total stiffness energy for each degree of freedom of joint motion is represented by a number (or eigenvalue); the magnitude of that number represents its contribution to forming the height of the bowl in that particular dimension. Eigenvalues less than zero indicate the potential for instability. The eigenvector (different from the eigenvalue) can then identify the mode in which the instability occurred, whereas a sensitivity analysis is used to reveal the possible contributors to unstable behavior. Gardner-Morse, Stokes, and Laible (1995) initiated interesting investigations into eigenvectors by predicting patterns of spine deformation due to impaired muscular intersegmental control. Their question was Which muscular pattern would have prevented the instability? Crisco and Panjabi (1992) began investigations into the contributions of the passive tissues. Our group has investigated the eigenvector by systematically adjusting the stiffness of each muscle and assessing stability in a variety of tasks and exercises. The contributions of individual muscles to stability are shown later in [figure 5.9](#).

Activating a group of muscle synergists and antagonists in the optimal way now becomes a critical issue. In clinical terms, the full complement of the stabilizing musculature must work harmoniously to ensure stability, the generation of the required moment, and the desired joint movement. But only one muscle with inappropriate activation amplitude may produce instability, or at least unstable behavior could result from inappropriate activation at lower applied loads.

Muscles Create Force and Stiffness

When a muscle contracts, it creates both force and stiffness. Although stiffness is always stabilizing, force may stabilize or destabilize. The relationship between increasing force in a muscle and the corresponding stiffness is quite nonlinear (Brown and McGill, 2010), given that large increases in stiffness occur quite early as activation begins (Brown and McGill, 2008). Thus, as a muscle becomes more active, it usually adds to spine stability; but if the force keeps rising, little additional stiffness is

created and the force of the muscle becomes large enough to actually induce buckling of the spine (Brown and McGill, 2005). This is more evidence for the wisdom of clinically enhancing spine stability with the objective of balance in stiffness and force in all contributing muscles rather than focusing on a single muscle group.

Sufficient Stability

How much stability is necessary? Obviously, insufficient stiffness renders the joint unstable, but too much stiffness and coactivation imposes massive load penalties on the joints and prevents motion. We can define sufficient stability as the muscular stiffness necessary for stability, with a modest amount of extra stability to form a margin of safety. Interestingly, given the rapid increase in joint stiffness with modest muscle force, large muscular forces are rarely required.

Cholewicki's work (Cholewicki and McGill, 1996; Cholewicki, Simons, and Radebold, 2000) demonstrated that, in most people with an undeviated spine, modest levels of coactivation of the paraspinal and abdominal wall muscles result in sufficient stability of the lumbar spine. This means that people, from patients to athletes, must be able to maintain sufficient stability in all activities—with low but continuous muscle activation. Thus, maintaining a stability margin of safety when performing tasks, particularly the tasks of daily living, is not compromised by insufficient strength but probably by insufficient endurance. We are now beginning to understand the mechanistic pathway of those studies, which show the efficacy of endurance training for the muscles that stabilize the spine. Having strong abdominal muscles does not necessarily provide the prophylactic effect that many hoped for. However, several works suggest that muscular endurance reduces the risk of back troubles (Biering-Sorensen, 1984). Finally, the Queensland group (e.g., Richardson et al., 1999) and several others (e.g., O'Sullivan, Twomey, and Allison, 1997; Silfies et al., 2009) noted the disturbances in the motor control system following injury (detailed in chapter 4). These disturbances are not primarily in one muscle but are widespread and compromise the ability to maintain sufficient stability. In summary, stability comes from stiffness, passive stiffness is lost with tissue damage, and active stiffness throughout the range of motion is lost with perturbed motor patterns

following injury.

Stability Myths, Facts, and Clinical Implications

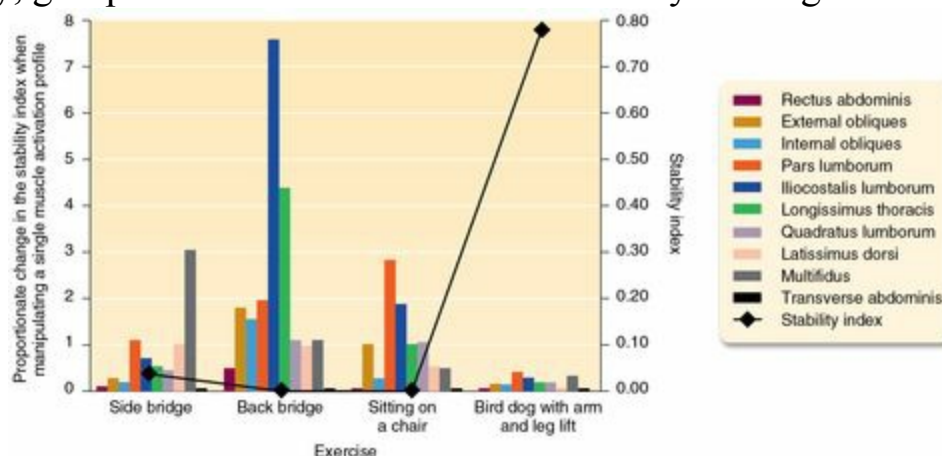
Having explored some of the issues of spine stability, we now consider a few crucial questions that will enhance clinical decisions.

- How much muscle activation is needed to ensure sufficient stability? The amount of muscle activation needed to ensure sufficient stability depends on the person and the task. For normal people walking, the normal muscular patterns are sufficient. A person in pain may need additional muscular stiffness to control the joint micromovements. Generally, for most tasks of daily living (e.g., holding and carrying loads), very modest levels of abdominal wall cocontraction (activation of about 10% of maximal voluntary contraction or even less) are sufficient. Again, depending on the task, cocontraction with the extensors (this also includes the quadratus) will ensure stability. However, if a joint has lost stiffness because of damage, more cocontraction and stiffness is needed to control pain-inducing joint micromovements. A specific example is shown in chapter 10.
- Is any single muscle most important? Several clinical groups have suggested focusing on one or two muscles to enhance stability. This would be similar to emphasizing a single guy wire in the fishing rod example. Rarely would it help; rather, it would be detrimental to achieving the balance in stiffness needed to ensure stability throughout the changing task demands. In particular, clinical groups have emphasized the multifidus and transverse abdominis. The Queensland group performed some of the original work emphasizing these two muscles. This was based on their research noting motor disturbances in these muscles in some people following injury. In fact, they developed a tissue damage model suggesting that chronically poor motor control (and motion patterns) initiates microtrauma in tissues that accumulates, leading to symptomatic injury (Richardson et al., 1999). Further, according to the Queensland model, injury leads to further deleterious changes in motor patterns such that chronicity can be broken only with specific techniques to reeducate the local muscle–motor control system. The intentions of the Queensland group were to address the documented motor deficits and attempt to reduce the risk of aberrant motor patterns

that could lead to the pathology-inducing patterns in their damage model. In other words, their recommendations appear to be directed toward reeducating the faulty motor patterns. However, many clinical groups have interpreted this approach to mean that these two muscles should be the specific targets when teaching stability maintenance over all sorts of tasks. This is problematic.

Hopefully, the understanding obtained from the stability explanation provided earlier will underscore the folly of this unidimensional emphasis. In fact, the multifidus or any other muscle can be the most important stabilizer at an instance in time. Four exercises are presented as an example (see [figure 5.9](#)) in which the contributors to stability are shown. The importance of individual muscles will change and adjust with subtle changes in relative muscle activation. In these examples the adjustments of all the laminae in each whole muscle are averaged to evaluate the entire muscle's role. Obviously, adjustments in a single laminae of some muscles would result in yet a different distribution of contributions to stability. Evaluating a wide variety of tasks and exercises is most revealing. The muscular and motor control system must satisfy requirements to sustain postures, create movements, brace against sudden motion or unexpected forces, build pressure, and assist challenged breathing, all the while ensuring sufficient stability. Virtually all muscles play a role in ensuring stability, but their importance at any point in time is determined by the unique combination of the demands just listed (Kavic and McGill, 2004). Every study that has actually quantified spine stability has reached the same conclusion—all muscles are important. This implies that a clinical focus on one muscle and not the whole system is detrimental. In some instances, the performance of one muscle enhances another (e.g., the rectus abdominis assists the obliques for stabilization). Further, evidence has demonstrated that endurance in these muscles is necessary to facilitate the continuous low-level stiffening contractions required for the maintenance of sufficient stability (e.g., Wagner et al., 2005).

Figure 5.9 Measuring individual muscle influence on lumbar spine stability during the performance of four stabilization exercises demonstrates that all muscles are important and that their relative importance changes with the task. Total stability is also indicated by the determinant, demonstrating how stability can change between tasks. (This is an example from a single subject. Typically, group means indicate a different stability ranking of exercises.)



Based on N. Kavcic, S. Grenier, and S.M. McGill, 2004, "Determining the stabilizing role of individual torso muscles during rehabilitation exercises," *Spine* 29(11): 1254-1265.

As noted elsewhere in this book, focus on a single muscle such as transverse abdominis (TvA) does not enhance stability or affect many people's back pain. There is a disconnect between the science and clinical practice regarding this muscle. Many clinicians have been taught to have all patients with back pain begin isolation training of TvA. Some patients may have very recognizable disturbances in motor patterns, yet these are ignored with an overzealous focus on TvA. For example, a patient may present with poor posture causing chronic muscle activity when loading the spine in compression. He may be bending painful discs because of his movement choices. Yet some clinicians fail to see these overt pain mechanisms and begin all back pain patients with isolated TvA training. Simple movement and posture training could have relieved the pain immediately. Perhaps even more disturbing is the number of high-performance coaches from around the world who instruct their athletes to draw their navels toward their spines to activate TvA, which they claim enhances stability while training or performing. Drawing the navel toward the spine reduces stability; increasing the distance between the abdominal wall and the spine enhances stability. A major question that must be asked is this: Is there even a disturbance in TvA?

In most people with back pain, I believe many other more important disturbances should be addressed. Efficacy studies support this view (e.g., Allison, 2012; Brooks, Kennedy, and Marshall, 2012; Hebert et al., 2010; Unsgaard-Tondel et al., 2012; Vasseljen et al., 2012; Wong et al., 2013).

Although much of the TvA focus was motivated by reports of onset delays in rapid arm movements, finding patients with a delayed TvA is very difficult. Several studies have failed to find any systematic TvA delay in patients with nonspecific back pain (Gubler et al., 2010). The paradigm to detect a delay is a rapid arm raise. Finding delays in tasks more common to daily living would be helpful in understanding the motor control and biomechanical importance. For example, other variables of motor control disturbances are very blatant and easily measured in patients. However, the clinical community does not seem to collectively appreciate that many muscles display delays in people with back pain and that the patterns of delay among different muscles are helpful in classifying the nature of the link between motor control and pain. Silfies and colleagues (2009) showed delays in several muscles and in fact could categorize subpopulations of pained patients based on this criterion.

It is possible that some patients benefit from specific TvA training. I suspect that these are the “abdominal sloths” who do not contract their abdominal wall muscles when defecating or when performing functional tasks such as lifting or carrying. TvA training would enhance their awareness of the existence of the abdominal wall, and I have seen this and appreciate the research. We prefer teaching bracing to encourage activation of all components of the abdominal wall; this immediately reduces pain in many patients (Ikeda and McGill, 2012). It is very stabilizing. Consider the heel drop test that initiates pain. Repeating with a tuned abdominal brace often eliminates pain. Obviously, the brace contraction is then tuned to optimize stability, reduce spine load, and reduce pain. The work of Koumantakis, Watson, and Oldham (2005) suggested that in the average painful back, specific TvA training delayed recovery until full abdominal wall work was undertaken. We suggest instead that patients simply perform stabilization training of the many muscles around the torso.

One final issue is whether the mechanics of TvA should be documented with ultrasound-derived data, as shown on EMG. Our data suggest that they should not. The link between TvA muscle thickening and activation is obscured by the composite structure of the three layers of the abdominal wall

(Brown and McGill, 2008, 2010) such that only EMG data can link activation with force and, with additional analysis, stiffness.

Some clinical issues remain and are addressed now:

- Do local and global or intrinsic and extrinsic stabilizers exist? From the classic definition and quantification of stability, the answer is no. Stability results from the stiffness at each joint in a particular degree of freedom. The relative contribution from every muscle source is dynamically changing depending on its need to contract for other purposes. The way the various contributors to stiffness add up, however, is important. In some instances, removing a muscle from the analysis has very little effect on joint stability. Once again, it depends on the demands and constraints unique to the task at that instant. The point is that all contributors are important for some tasks and should be recognized as potentially important in any prevention or rehabilitation program. Conversely, conceiving of stabilizers as intrinsic or extrinsic may offer no benefit for clinical decision making.
- What are stabilization exercises? Stabilization exercises and a stable core are often discussed in exercise forums. What are stabilization exercises? The fact is that any exercise can be a stabilization exercise depending on how it is performed. Sufficient joint stiffness is achieved through the creation of specific motor patterns. To adapt a phrase popular in motor control circles, Practice does not make perfect; it makes permanent. Ideally, good stabilization exercises that are performed properly groove motor and motion patterns that ensure stability while satisfying all other demands. Thus, an exercise repeated in a way that grooves motor patterns and ensures a stable spine constitutes a stabilization exercise (McGill, 2001). However, some stabilization exercises are better than others; again, it depends on the objectives. For example, the resultant load on the spine is rarely considered. One key to improving painful backs is to select stabilization exercises that impose the lowest load on the damaged spine. Chapter 4 provides a ranking of such exercises. The big three exercises have been shown to create a lasting stiffness and stability (Lee and McGill, 2014a, 2014b) in a spine-sparing way. On the other hand, the best stabilization exercise for a high-performance athlete will involve the establishment of dynamic and complex motion patterns, all the while ensuring sufficient

spine stability. An important requirement for many athletes is to maintain a stable spine while breathing hard, such as when playing high-intensity sports. This is discussed in chapter 11.

- Can clinicians identify those who are poor stabilizers? From a qualitative perspective, the answer is yes, as long as the clinician is cued into seeing and feeling compromised motion and muscle activation patterns. From a quantitative perspective, we have used modeling analysis to find ways to identify those who compromise their lumbar stability via specific motor control errors. We observed such inappropriate muscle sequencing in men who were challenged by holding a load in the hands while breathing 10% CO₂ to elevate breathing. On one hand, the muscles must cocontract to ensure sufficient spine stability; but on the other, challenged breathing is often characterized by a rhythmic contraction–relaxation of the abdominal wall (McGill, Sharratt, and Seguin, 1995). Thus, the motor system is presented with a conflict: Should the torso muscles remain active isometrically to maintain spine stability, or will they rhythmically relax and contract to assist with active expiration (but sacrifice spine stability)? Fit motor systems appear to meet the simultaneous breathing and spine support challenge; unfit ones may not (Grenier and McGill, 2007, 2008; Wang and McGill, 2008). All of these deficient motor control mechanisms heighten biomechanical susceptibility to injury or reinjury (Cholewicki and McGill, 1996). We are currently using this paradigm in a longitudinal study to see whether those who border on instability during this challenge will be the ones who develop low back troubles. Some other tests to detect functional instability are described in chapter 9.

A Final Note

In summary, achieving stability is not just a matter of activating a few targeted muscles, be they the multifidus, transverse abdominis, or any other. Sufficient stability is a moving target that continually changes as a function of the three-dimensional torques needed to support postures. It involves achieving the stiffness needed to endure unexpected loads, preparing for moving quickly, and ensuring sufficient stiffness in any degree of freedom of the joint that may be compromised from injury. Thus, this stiffness is tuned to the situation with a skilled level of coordinated muscular cocontraction. Motor control fitness is essential for achieving the stability target under all possible conditions for quality performance and injury avoidance. Finally, the terms stability and stabilization exercise have been used by many to mean different things. When one trial reports no efficacy for stabilization exercise, beware, read carefully, and determine whether the exercises were truly consistent with bonafide stability.